

Problem Set 3

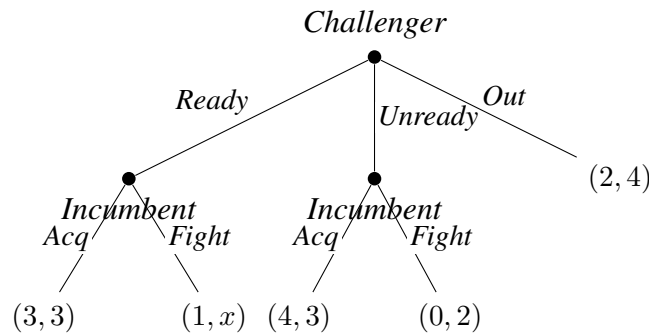
POLS W4732

Fall 2026

Due October 16 before the TA section

Artificial-intelligence policy. Unless an exercise explicitly states otherwise, students may not use generative-AI systems (including ChatGPT, Claude, Gemini, Copilot, or AI-generated search answers) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, typesetting software, the teaching team, and collaboration within the registered problem-set group are permitted. Any approved computational tool must be disclosed. Every group member is responsible for understanding and being able to explain the entire submission.

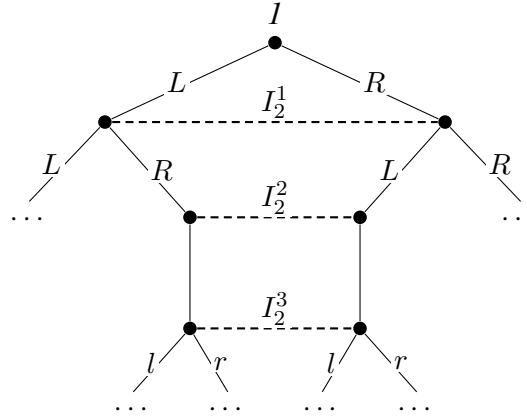
Exercise 1 (Entry and strategic-form representation). *Consider the game below.*



The incumbent payoff at (Ready, Fight) is the parameter x .

1. Construct the strategic form and list all Nash equilibria. For each, state its induced path and the off-path action that supports it.
2. Characterize the values of x at which the Nash and SPE sets change.
3. A proposed solution deletes an incumbent action because its information set is not reached. Explain why that deletion is or is not legitimate in strategic-form Nash equilibrium.

Exercise 2 (Mixed and behavioral strategies). *In the game below, the dashed lines are Player 2's information sets. Player 2 has pure strategies $s_2 = (L, l, r)$ and $s'_2 = (R, r, l)$ and mixes equally between them.*



Student A claims that an equivalent behavioral strategy assigns probability $1/2$ to each action at every information set. Student B claims that no realization-equivalent behavioral strategy exists. Calculate the probability of every terminal history affected by Player 2's choices, determine which claim is correct, and identify whether the game satisfies perfect recall. Modify the smallest number of information-set links needed to restore perfect recall and repeat the equivalence calculation.

Exercise 3 (Public-good provision). Each of n players privately observes a cost c_i drawn independently from a continuously differentiable distribution F on $[\underline{c}, \bar{c}]$. Each chooses whether to provide a public good. Every player receives benefit 1 if at least one player provides it; a provider also pays c_i .

1. Derive the symmetric cutoff equation and conditions sufficient for uniqueness.
2. Show how the density at the fixed point affects the response of the cutoff to a location shift in costs.
3. Specialize to the uniform distribution $[\gamma - (2\psi)^{-1}, \gamma + (2\psi)^{-1}]$ and recover the two-player comparative statics.
4. For n players, distinguish the cutoff, an individual's contribution probability, and the probability that at least one player contributes.
5. Construct two distributions with the same mean for which these objects move differently, or prove this cannot occur under additional assumptions that you state.

Exercise 4 (Incomplete-information war). Countries 1 and 2 choose F (mobilize) or N (accommodate). Mobilizing costs μ . If both mobilize, the winner receives $1 - \mu$ and the loser $-1 - \mu$; choosing F against N gives $-\mu$, and choosing N gives zero. Nature draws $\omega \in \{UR, DR, UL, DL\}$; Country 1 observes the first letter and Country 2 the second. Country 1 wins in UR and DL .

The common prior satisfies $\Pr(UR) = \Pr(DL) = q/2$ and $\Pr(UL) = \Pr(DR) = (1 - q)/2$, with $q \in (0, 1)$ and $\mu \in (0, 1)$.

1. Construct a table of each type's posterior probability of winning.
2. Characterize all pure-strategy Bayesian equilibria as a function of (q, μ) .
3. For every excluded candidate equilibrium, give one type and one profitable deviation. Interpret the benchmark $q = 1/2$.